

Design and Verification of Circuit Topologies Involving Bipolar Junction Transistors

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Abstract—Bipolar Junction Transistors (BJTs) are important components with a wide variety of applications, including amplification, digital logic, and filtering. This report will follow a series of experiments performed to verify the different operating behaviour of BJT circuits with an emphasis on amplification. The circuits described will only contain one BJT, and will demonstrate certain characteristics of these components. Finally, the report will conclude with the construction of an amplifier circuit using a single BJT.

I. INTRODUCTION

BJTs are semiconductor devices built by alternating three doped semiconductors to create two junctions. There are two ways to design BJTs, both of which result in different components. An n-type semiconductor can be placed next to a p-type material, and then another n-type can be placed next to the p-type. This is called an NPN transistor. The opposite can be done as well, resulting in a PNP transistor. Both have similar behaviour, but this report will focus on the NPN BJT.

A. The NPN Transistor

Both types of BJTs have three terminals, defined by each of the doped semiconductors. As can be seen in figure 1, the terminals are called the collector (C), the emitter (E) and the base (B). The voltage differences between these terminals defines what operating mode the transistor is in. Each mode has different behaviour and the experiments presented here will focus solely on the forward-active mode. This mode is defined to be when the voltage difference from the collector to the base is positive, or that the base voltage is 'less' than the collector voltage.

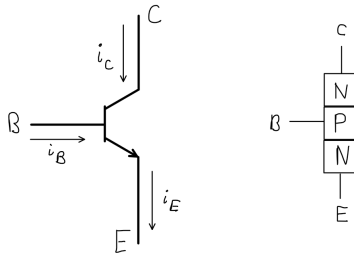


Fig. 1. basic diagram of NPN BJT device, with the circuit symbol (left) and the semiconductor arrangement (right), showing the different terminals

In Forward-Active mode, there are approximate relations between the various currents and voltages, that are valid only in this mode [5].

$$i_C \approx I_S \left(1 + \frac{v_{CE}}{V_A} \right) \exp\left(\frac{v_{BE}}{V_T}\right) \quad (1)$$

$$i_B = \frac{i_C}{\beta_F} \quad (2)$$

$$i_E = -i_C - i_B \quad (3)$$

The values of β_F , V_A , and I_S are characteristic of the specific device, called the current gain, early voltage, and saturation current respectively. In addition, V_A is usually a very large value, $V_A \gg v_{CE}$, so equation (1) can typically be approximated as [5]:

$$i_C \approx I_S \exp\left(\frac{v_{BE}}{V_T}\right) \quad (4)$$

The voltage V_T is the thermal voltage, which is $V_T \approx 26$ mV at room temperature. The following sections in this report will verify these relationships using experimental evidence.

B. Amplification Topologies

As stated previously, a major applications of these types of transistors is current amplification. The common-emitter amplifier (CE) is an example topology that will be tested in this experiment. It involves inputting a current to the base terminal and taking the output at the collector [2]. The typical input current is on the order of single microamps (μA), which is useful in applications using small signals.

The design for one of these amplifiers is shown in figure 2. The voltage gain from the figure 2 circuit can be calculated using a quantity called the transconductance g_m . This is a value that characterizes the amplifier, and the gain is then [5]:

$$G = -g_m R_C \quad (5)$$

Where R_C is the resistor attached to the collector, and g_m is calculated based on the collector current:

$$g_m = \frac{I_C}{V_T} \quad (6)$$

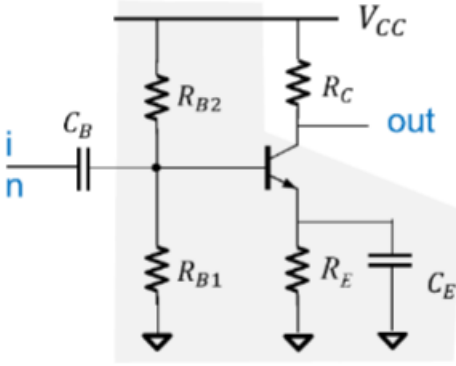


Fig. 2. common-emitter amplifier design using a capacitor to bypass the emitter and achieve large gain [5]

II. EXPERIMENTAL PROCEDURES

This section will discuss the various experiments performed to verify the behaviour of the transistor in certain situations. The transistor model used is BC547 [1], which is a three terminal NPN transistor. A variety of circuits were constructed, first to extract certain model parameters, and then to test different amplifier topologies.

A. Extraction of the Current Gain

The first circuit was constructed to measure the current gain β_F for the transistor. The design is shown in figure 3, and the measured values are shown in table 1. The resistor used at the emitter terminal was a 3.3 k Ω (measured at 3.302 k Ω), and the collector voltage was set at a constant 10 V.

To find the current gain, the relationship between the collector current and the base current is needed. The base current can be measured as that is the input, and the emitter voltage can be measured using a digital multimeter [3]. From the emitter voltage, the current I_E can be calculated, and then the collector current can be calculated using equation (3). Once the collector current for each base current is found, equation (2) will provide the value for the current gain.

As can be seen, the behaviour is not very determinate at low base voltages, but the average current gain was $\beta_F = 587.5$ which is consistent with the datasheet [1].

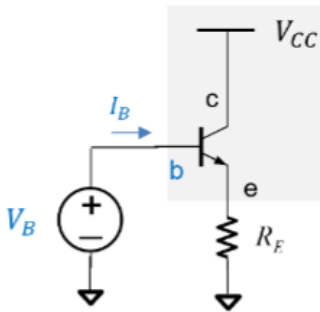


Fig. 3. simple circuit designed to measure the current gain of the transistor device. The collector terminal is biased with a constant DC source and the base terminal is the input [5]

TABLE I
MEASURED VALUES FOR I_B AND V_E AT DIFFERENT VALUES OF V_B , AND RESULTING CALCULATED VALUES OF I_E , I_C , AND β_F

V_B [V]	I_B [μ A]	V_E [V]	I_E [μ A]	I_C [μ A]	β_F
0.5	0	0	3.93	3.93	N/A
1	0.222	0.422	127	127.2	627.7
1.5	0.472	0.902	273	273.5	577.4
2	0.728	1.391	421	421.7	578.1
2.5	0.986	1.883	570	571	577.6
3	1.246	2.377	718	719.2	576.7

B. Constant Current Sink

Since the collector current is only dependent on the base current (to a good approximation, see equation 4), biasing the base at a constant voltage should force the collector current to also be constant as well. Thus the collector voltage can be varied without changing the current. In this form, the transistor behaves as a constant current sink [5].

The circuit to verify this behaviour is shown in figure 4. The base terminal is biased using a resistive voltage divider involving R_{B2} and R_{B1} . The values of the resistors were chosen to be $R_{B1} = 47$ k Ω and $R_{B2} = 200$ k Ω (measured 46.74 k Ω and 199.3 k Ω), meaning the base terminal voltage was about 1.9 V (as $V_{CC} = 10$ V). This was found using the well known voltage divider equation [2]. As seen previously, the resistor attached to the emitter terminal was 3.3 k Ω .

The collector voltage was stepped from 2 V to 10 V and the collector current was measured. The results are shown in table 2.

TABLE II
MEASURED VALUES OF I_C FOR DIFFERENT VALUES OF V_C SHOWING THE TRANSISTOR BEING USED AS A CONSTANT CURRENT SINK

V_C [V]	1.5	2.0	4.0	6.0	8.0	10.0
I_C [μ A]	151.4	382.9	383.3	383.6	383.9	384.9

As can be seen from the data, the transistor clearly behaves as a constant current sink, as I_C did not change at all over a wide range of voltages. For experimental purposes, a measurement was taken with $V_C = 1.5$ V, but at that voltage the collector current was measured at about 150 μ A. This

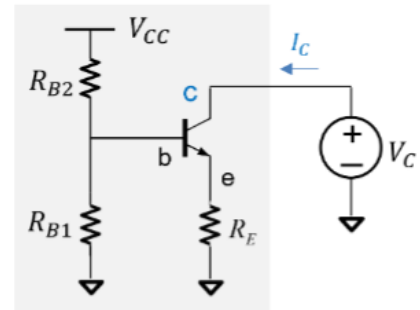


Fig. 4. circuit that uses the transistor as a constant current sink by biasing the base terminal at a constant voltage (and therefore constant current). The collector voltage is stepped through different values to test this [5]

seems to contradict the constant current sink, but with this collector voltage the transistor is no longer in forward active mode, and is instead in saturation, since $V_C < V_B$. This means the equations 1-4 are no longer valid to describe the operation of device, accounting for the discrepancy in measurement. Thus only the measurements at 2 V and above are valid for this experiment, showing that the transistor was indeed a constant current sink.

C. Common-Emitter Amplifiers

The final experiment performed involved the transistor in an amplifier topology. Specifically, a common-emitter amplifier as discussed in section 1.B, but simpler to start. The circuit constructed is shown in figure 5.

Understanding the operation of this device requires some theoretical work. The transcendental equation describing the operation of this circuit is as follows [5]:

$$V_{Th} \approx \left(\frac{R_{Th}}{\beta_F} + R_E \right) I_C + V_T \ln \left(\frac{I_C}{I_S} \right) \quad (7)$$

All values in this equation are known except for I_C , which cannot be solved for algebraically. The thevenin resistance is based on the divider involving R_{B1} and R_{B2} which are the same values as the previous section, which gives a thevenin resistance of 38.1 k Ω . Similarly, the thevenin voltage is 1.903 V. The method to find the collector current involves rearranging the equation so that each side is a function of I_C , and then finding where the functions intersect. This process is shown in figure 6, and the resulting values is $I_C \approx 377 \mu\text{A}$.

After finding the collector current, the transconductance can be calculated using equation (6), and the gain calculated as well. The transconductance was found to be $g_m \approx 13.31 \text{ mA/V}$. The gain for the circuit shown in figure 5 can be calculated from [5]:

$$G = - \frac{g_m R_C}{1 + g_m R_E} \quad (8)$$

The R_C resistance was chosen as 10 k Ω (measured 9.904 k Ω), meaning the voltage gain would be calculated to be $G \approx -2.96 \text{ mV/mV}$ for this common-emitter amplifier.

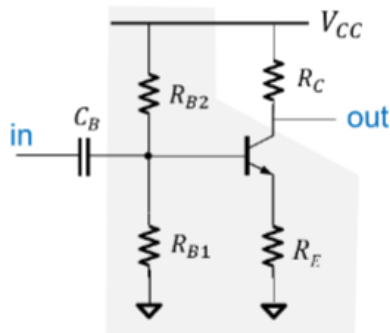


Fig. 5. a basic common-emitter amplifier in which the collector current is allowed to vary due to the addition of a resistor at the collector terminal [5]

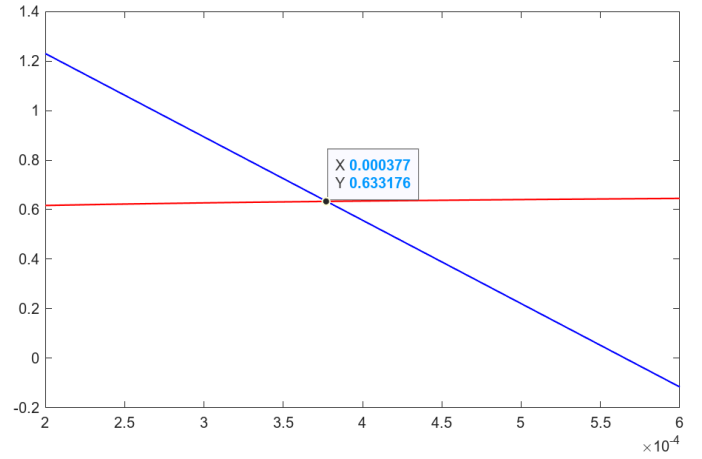


Fig. 6. solving the transcendental equation (7) by finding the intersection point of both sides of the equation

To test this theory, a sine wave with $V_{pp} = 100 \text{ mV}$ and a frequency of 10 kHz was applied to the input of the circuit in figure 5, which was coupled to the base terminal using a capacitor of value $C_B = 10 \text{ nF}$ (measured 9.1 nF). Figure 7 demonstrates that this capacitor did not effect the input waveform.

The output was then taken as the voltage at the collector, which is shown in figure 8. The RMS values were measured using the oscilloscope[4]. The input signal was measured to have an RMS of 70.25 mV and the output was measured with an RMS of 203.8 mV. The experimental gain was then 2.901 mV/mV. This has about a 2% difference from the theoretical value predicted by equation (8). In addition, the scope capture in figure 8 shows that the output signal is perfectly out of phase with the input signal, meaning the gain is negative which is consistent with the theoretical predictions.

A gain of 3 is smaller than we would like from these resistor values, as increasing the gain would require a very large R_C . Fortunately, a modification can be made to the figure 5 circuit

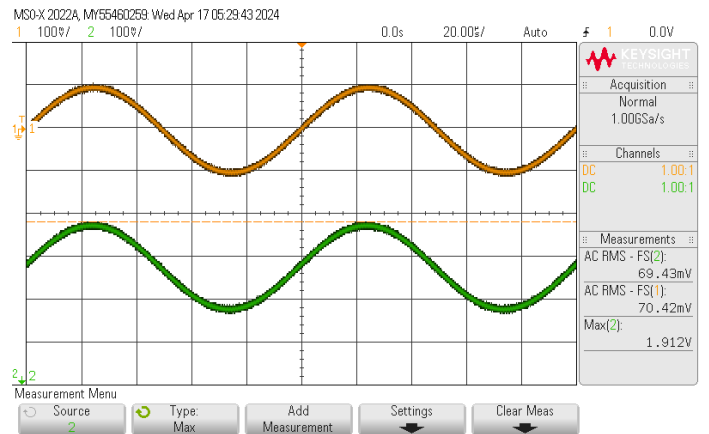


Fig. 7. scope[4] capture showing the input signal to figure 5 (yellow) as well as the signal at the base terminal of the transistor (green). This shows how the capacitor C_B does not effect the input signal

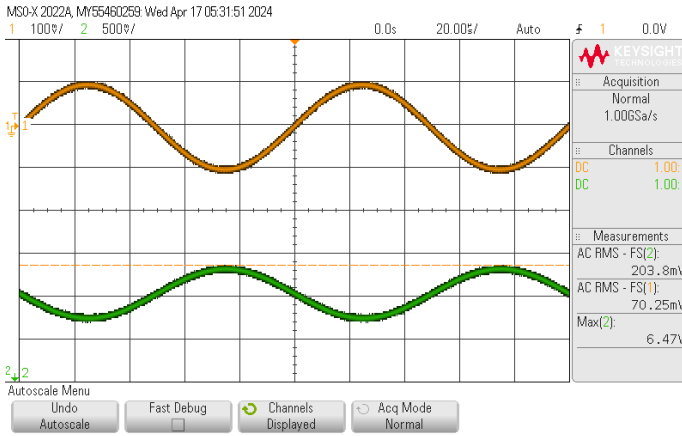


Fig. 8. input (yellow) versus output (green) of the circuit in figure 5. The scales on the waveforms are different, with the output curve being depicted at a larger scale

to force the gain to be higher. Bypassing the emitter resistor with a large value capacitor will cause the emitter voltage to hold constant. This means the entire input voltage will be forced across the base emitter junction, increasing the collector current by a significant amount. This is the circuit seen in figure 2, and the gain is described by equation (6).

From this equation, and the transconductance calculated earlier, the theoretical gain is -133.1 mV/mV . The verification of this circuit is shown in figure 9. In order to keep the various currents within the safe range [1], a voltage divider (200 k Ω and 10 k Ω) was used to significantly reduce the input voltage. The bypass capacitor was chosen as a 330 μF (measured 323 μF) capacitor which acts as an AC short at the input frequency.

The input signal appeared to have a lot of noise, which was due to the small magnitude of the waveform. The input peak-to-peak voltage was divided down to around 4 mV, which is close to the amount of noise that was present in the room. Therefore the noise is a prominent part of the waveform and

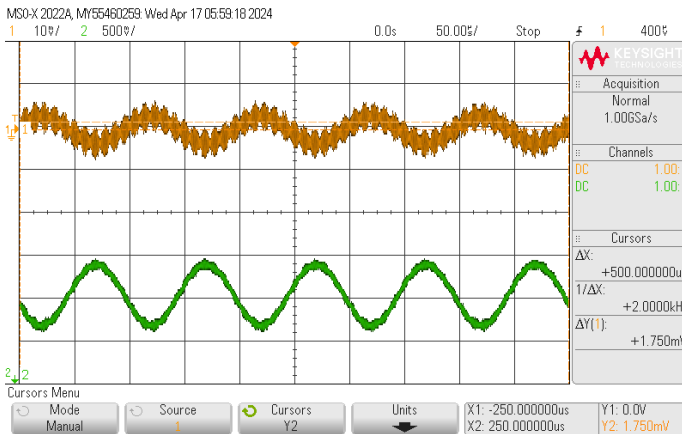


Fig. 9. scope[4] capture showing the input (yellow) and output (green) signals of the figure 2 common-emitter amplifier. The input signal shows significant interference from noise due to the incredibly small peak-to-peak voltage, and the output shows a significant gain from the input

affected the measurements made by the scope.

Using cursors on the scope provided a more accurate way to measure the RMS. Using the cursors the input RMS was measured to be 1.75 mV, and the output RMS was measured to be 265 mV. This means the experimental gain factor was 151 mV/mV, which is a 14% difference from the theoretical prediction. The input and output are perfectly out of phase, showing that the gain is negative, as is predicted by theory.

III. CONCLUSION

Bipolar Junction Transistors are a line of semiconductor devices used in a wide variety of applications in the electrical world. The main two are amplification and switching, for use in signal processing and digital logic, integral parts of electrical projects. Using an NPN BJT [5], the various current and voltage characteristics of BJTs were verified through experimental means.

The current gain, β_{AF} , represents how much the the current gain is between the base and collector terminals, and is a property of the specific device. In section II.A this parameter was determined experimentally using measurements of the collector current and various input base currents.

In section II.B, the transistor was demonstrated to be a constant current sink under certain operating conditions, specifically when the base voltage is held constant. Finally, signal amplification was explored in section II.C, demonstrating use of the transistor as a voltage amplifier, and comparing similar topologies by measuring the gain. It was found that bypassing the emitter terminal using a capacitor to hold the voltage constant would significantly increase the gain of the amplifier, a useful tool for signal processing.

To conclude, the BJT is an interesting and complex circuit that is useful in many different circuits, demonstrating the benefit of utilizing semiconductor devices in circuitry across all applications.

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